

Conformal Prediction for Protein Stability

Protein-Level Calibration and Sign Reliability

Guido Barducci

University of Turin

Protein stability and mutation effects

Mutations can change the free-energy difference between folded and unfolded states.

- Stability changes matter for protein engineering and variant interpretation.
- We predict measured $\Delta\Delta G$ values in kcal mol^{-1} .

Sign convention in this talk	Interpretation
$\Delta\Delta G > 0$	Stabilizing
$\Delta\Delta G < 0$	Destabilizing

Barducci, manuscript, Introduction. Some tools use the opposite sign convention.

JanusDDG: sequence-based prediction

JanusDDG uses protein language-model embeddings and attention between WT and mutant sequences.

- Sequence-based prediction of mutation-induced stability changes.
- Thermodynamic consistency motivates the model design.

$$\widehat{\Delta\Delta G}(A, B) = -\widehat{\Delta\Delta G}(B, A).$$

This study builds on JanusDDG to model predictive uncertainty.

Barducci et al., *Communications Biology* 9, 494 (2026). DOI in the reference slide.

Uncertainty for individual predictions

MAE and correlation summarize performance across a dataset. An interval provides a range of plausible measured $\Delta\Delta G$ values for a particular mutation.

- BayeStab estimates predictive uncertainty with a Bayesian neural network.
- FoldX error regression produces mutation-specific prediction intervals.

Our focus: conformal calibration with equal protein weights.

Wang et al., *Protein Science* 31, e4467 (2022).

Sapozhnikov et al., *BMC Bioinformatics* 24, 426 (2023).

Conformal prediction: residual calibration

Illustration with ordinary split conformal prediction

Held-out calibration errors determine the interval radius:

$$s_i = |y_i - \hat{f}(x_i)|, \quad C_{1-\alpha}(x) = [\hat{f}(x) - q, \hat{f}(x) + q].$$

Here q is the finite-sample-corrected calibration quantile.

- Exchangeability supports marginal coverage of at least $1 - \alpha$.
- The guarantee concerns a future observation drawn from the target population.

Angelopoulos, Barber and Bates, *Theoretical Foundations of Conformal Prediction*, arXiv:2411.11824v5 (2026).
Coverage statement: ordinary split conformal.

Equal calibration weight for each protein

Proteins contribute unequal numbers of measured mutations.

Toy example	Mutations	Uniform mutation mass	Our mass
Protein A	100	90.9%	50%
Protein B	10	9.1%	50%

$$w_i = \frac{1}{P n_{p(i)}}, \quad \sum_{i \in p} w_i = \frac{1}{P}.$$

Target: sample a protein first, then a mutation within it.

Illustrative counts. Calibration weights follow manuscript Section 2.6.

ProbJanusDDG: mean and local scale

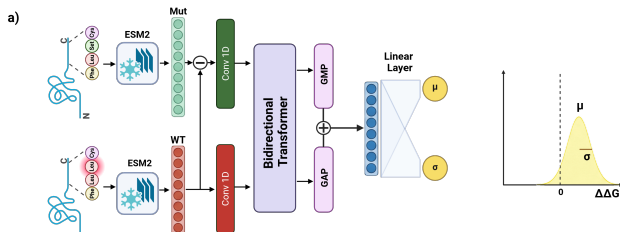
Antisymmetric mean

$$\mu(A, B) = -\mu(B, A).$$

Symmetric scale

$$\sigma(A, B) = \sigma(B, A).$$

Positive scale modulates interval width.



Manuscript, Figure 1a and Section 2.3.

Protein-weighted CV+ calibration

Each calibration mutation uses a model that excluded its entire outer fold.

$$r_i = \frac{|y_i - \mu_{k(i)}(x_i)|}{\sigma_{k(i)}(x_i)}.$$

For a new mutation, preserve the score–model pairing:

$$\ell_i(x) = \mu_{k(i)}(x) - r_i \sigma_{k(i)}(x), \quad u_i(x) = \mu_{k(i)}(x) + r_i \sigma_{k(i)}(x).$$

$$C_{1-\alpha}(x) = [Q_{\alpha}^w(\ell_i(x)), Q_{1-\alpha}^w(u_i(x))].$$

Manuscript Section 2.6. Empirical weighted-quantile convention. Standard CV+ uses unnormalized absolute residuals with the same models and weights.

Training and external evaluation

Dataset	Mutations	Proteins	Role
S2450	2,450	115	Training / calibration
S669L	669	87	External evaluation
S461L	461	46	Curated, overlapping subset

- Five outer folds separate proteins using a 25% sequence-identity threshold.
- Epoch selection excludes the outer fold; refitting uses the other four folds.
- External targets enter evaluation only.

Manuscript, Sections 2.2 and 2.5. S461L does not provide independent replication.

External point-prediction performance

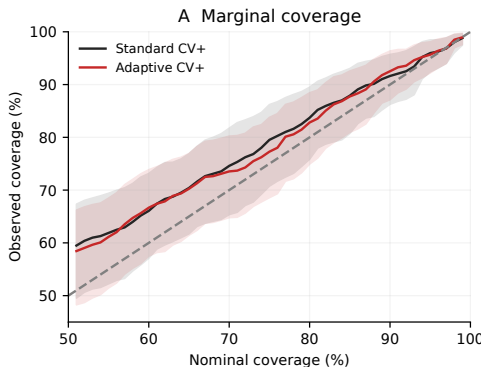
Dataset	Pearson r	Spearman ρ	MAE	MSE
S669L	0.53	0.55	0.99	2.03
S461L	0.70	0.68	0.74	0.97

MAE in kcal mol^{-1} ; MSE in $(\text{kcal mol}^{-1})^2$. All test mutations contribute equally to these summaries.

Point accuracy provides context for evaluating interval usefulness.

Manuscript, Table 1. Median of five fold-specific predictions.

Mutation and protein coverage



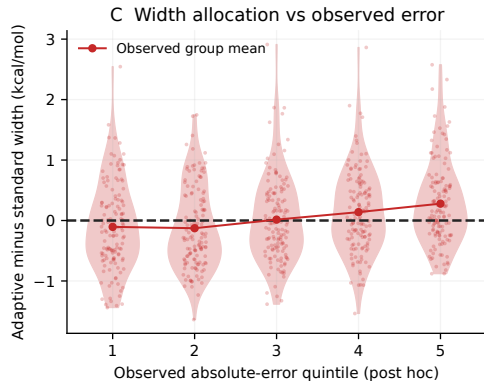
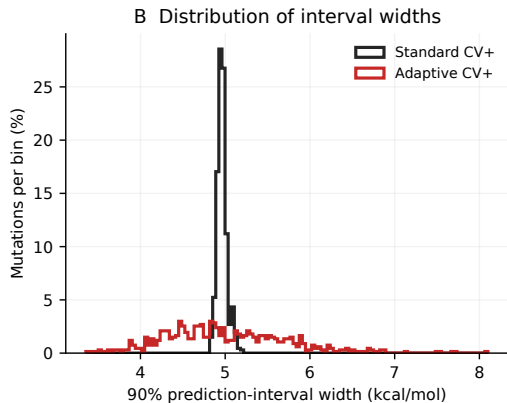
S669L, 90% nominal level

Coverage	Standard	Adaptive
Mutations	91.63%	92.53%
Proteins	86.76%	86.87%

Protein-averaged coverage: approximately 87%.

S669L. Figure 2a and Table 3. Curve and pointwise bootstrap bands weight mutations equally.

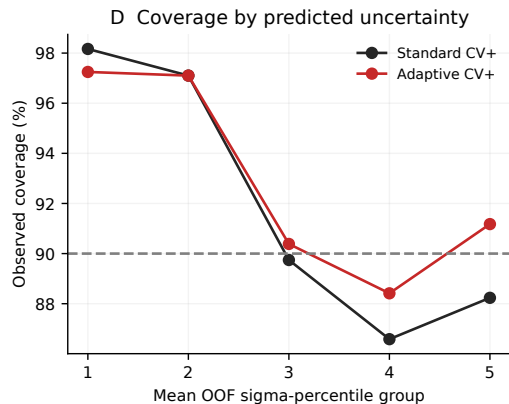
Adaptive intervals redistribute width



Mean full width: **4.964** standard vs **5.004** adaptive kcal mol⁻¹.

S669L, 90% nominal. Figure 2b–c and Table 3. Error quintiles are post-hoc diagnostics.

Learned scale and uncertainty groups



Error association

Spearman $\rho = 0.22$.

Coverage dispersion

Standard: 4.75 pp.

Adaptive: 3.63 pp.

S669L. Figure 2d and Table 3. Fixed OOF scale-percentile groups. No subgroup-coverage guarantee.

Sign-call frequency and accuracy

Stabilizing if $L_i > 0$; destabilizing if $U_i < 0$. Intervals touching zero yield no call.

Nominal	Calls	Call rate	Correct / called	Proteins reached
50%	229	34.23%	217/229 (94.76%)	46/87
60%	171	25.56%	167/171 (97.66%)	37/87
70%	104	15.55%	103/104 (99.04%)	21/87
90%	4	0.60%	4/4 (100%)	2/87

Higher nominal levels yield fewer directional calls.

S669L, adaptive CV+, manuscript Figure 4 and Table 5. Empirical accuracy among selected calls is distinct from nominal interval coverage. All four 90% calls are destabilizing.

Conclusions

- Protein-weighted calibration gives each observed protein equal influence.
- Adaptive CV+ redistributes interval width with a similar mean width.
- Zero-excluding intervals support selective sign decisions.

Scope of the evidence

Protein-averaged coverage remains below 90% on S669L.
Benchmark composition affects observed coverage.

Uncertainty evaluation complements point-prediction performance.

Backup: the curated S461L subset

90% nominal level

S461L metric	Standard	Adaptive
Mutation coverage	97.61%	98.05%
Protein-averaged coverage	96.56%	93.85%
Mean full width (kcal mol^{-1})	4.965	4.967
Mean interval score (kcal mol^{-1})	5.188	5.109

Benchmark curation changes observed calibration behavior.

Manuscript, Table 3. S461L overlaps S669L and does not provide independent replication.

Backup: calibration of a one-hot baseline

S669L, 90% nominal level

Adaptive intervals	ProbJanusDDG	One-hot baseline
Pearson r	0.53	0.33
MAE (kcal mol ⁻¹)	0.99	1.23
Mutation coverage	92.53%	90.28%
Protein-averaged coverage	86.87%	86.41%
Mean full width (kcal mol ⁻¹)	5.004	5.84

Lower point accuracy coincides with wider intervals.

Manuscript, Tables 1, 3 and 4, Section 3.3.5. Same training and calibration protocol, different architecture and representation.

Backup: coverage assumptions

- Independent protein clusters with aligned within-protein sampling.
- Data-independent fold allocation and fitting outside each calibration fold.

For the empirical weighted rule, the draft Appendix gives:

$$\Pr\{Y_* \in C_{1-\alpha}(X_*)\} \geq 1 - 2\alpha - \frac{2(1-\alpha)K}{P+K}.$$

Marginal bound for the protein-balanced target. Homology-based folds require separate justification.

Draft theoretical Appendix developed alongside the manuscript. The bound is not an assertion of exact nominal coverage or selected-sign accuracy.

Selected references

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